# Use-Case Flow Specification

**UC-01: Apply for Micro-Loan & Assess Credit Risk**

Student: Mudit Chaturvedi (PES1UG24CS279) | Problem Statement #33

**Use Case ID:** UC-01 (includes UC-02, UC-03, UC-04) | **Primary Actor:** Borrower | **Secondary Actors:** Credit Engine, Gateway

**Goal in Context:** Borrower submits loan request and financial metrics; system performs algorithmic risk scoring, generates an amortization schedule, and publishes to the peer marketplace.

## Preconditions & Postconditions

• **Preconditions:** (1) Borrower is registered with verified phone/email. (2) No delinquent active default notices. (3) Credit Engine service is operational.

• **Success Postconditions:** (1) Credit score (300-850) and risk tier (Low/Med/High) recorded. (2) Amortization schedule generated. (3) Loan published to marketplace as OPEN_FOR_FUNDING.

• **Failure Postconditions:** (1) No listing published. (2) Application tagged REJECTED with actionable feedback.

## Main Success Scenario (MSS)

1. **Application Initiation:** Borrower specifies requested principal (Rs. 40,000) and tenure (6 months).

2. **Financial Ingestion («include» UC-02):** Borrower submits monthly net income, existing debt obligations, and employment tenure.

3. **Heuristic Scoring («include» UC-03):** System computes Debt-to-Income (DTI = 21.5%), evaluates repayment history ratio (0.96), and calculates composite credit score (717/850).

4. **Risk Tiering:** System maps score to **Medium Risk** (benchmark APR: 15.0%, borrowing ceiling: Rs. 50,000).

5. **Amortization Generation («include» UC-04):** System produces reducing-balance schedule (Monthly EMI: Rs. 6,961.35).

6. **Borrower Acceptance:** Borrower reviews loan terms and accepts the offer digitally.

7. **Marketplace Activation:** Listing state set to OPEN_FOR_FUNDING; notification dispatched to peer lenders.

## Alternate Flows

• **A1: High-Risk Classification:** If credit score is 450–600 or DTI 45%–60%, system categorizes loan as High Risk (APR: 22.0%, cap: Rs. 15,000). If requested principal exceeds Rs. 15,000, borrower is prompted to reduce principal.

• **A2: Immediate Rejection (Excessive DTI > 60%):** If DTI > 60% or score < 450, application is set to REJECTED_HIGH_RISK with a 90-day cooling advisory. Use case terminates.

• **E1: Database Persistence Timeout:** If transaction fails during marketplace save, database rolls back immediately and borrower is advised to retry.